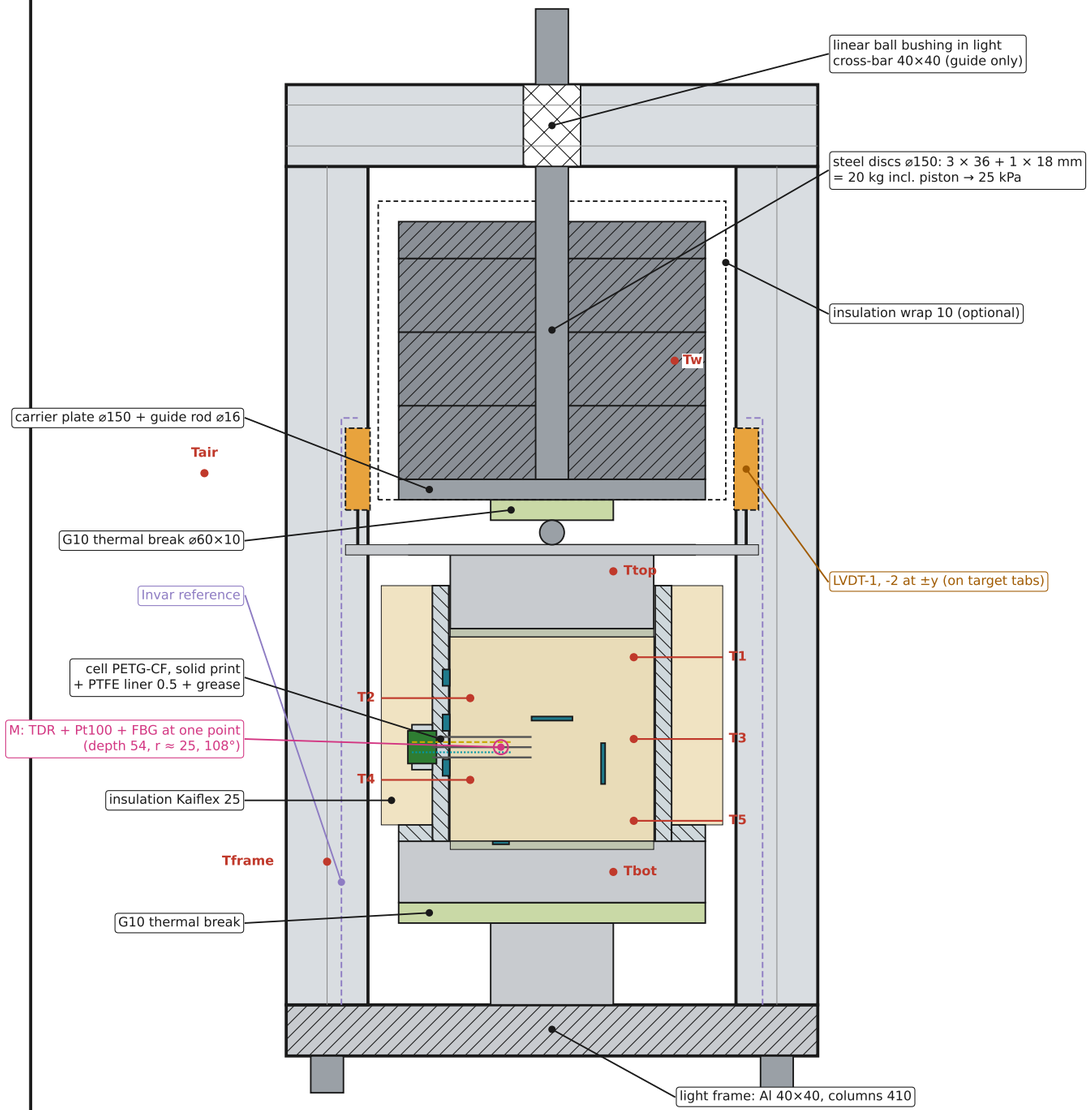


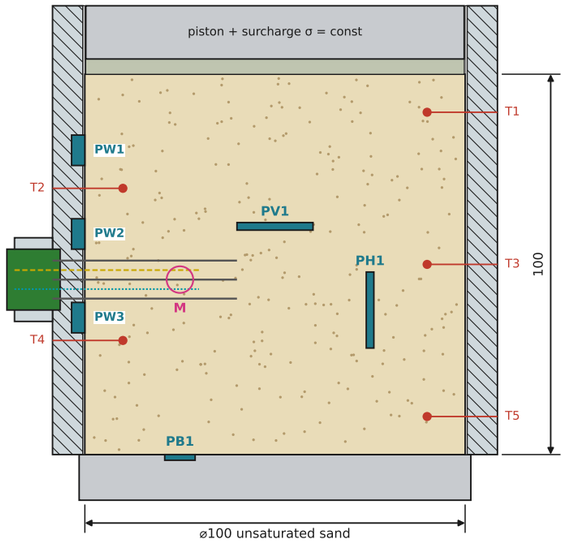

## Sensors (tag → model → CR1000)

| tag | measures | recommended model | CR1000 |
|---|---|---|---|
| PV1 / PH1 | vert. / radial stress | TML PDA-PA 200 kPa on ø20×2 disc | MUX-1 (4×16), BrFull, VX3 2 V |
| PB1 | vertical stress at base | TML PDA-PA 200 kPa (flush) | MUX-1 (4×16), BrFull, VX3 2 V |
| PW1…PW3 | lateral stress on wall | TML PDA-PA 200 kPa (flush) | MUX-1 (4×16), BrFull, VX3 2 V |
| LVDT-1,2 | check: no movement | RDP DCTH400 | SE1–2, VoltSE ±5 V |
| T1…T5 | soil, r 40 mm | Campbell 109SS | MUX-2 → SE13, VX2 |
| M: TDR | water / unfrozen water | mini 3-rod TDR 40 mm [U] + TDR200 | SDM C1–C3 |
| M: Pt100 | reference temperature | Pt100 1/10 DIN, 4-wire, ø3 | MUX-1 + 4WPB100, BrHalf4W |
| M: FBG | optical temperature | FBG temp. sensor in capillary ø1 [U] | PC (Ethernet), time-synced |
| Ttop, Tbot, Tw | piston, pedestal, weights | Campbell 109SS | (same) |
| Tframe, Tair | frame, air | Campbell 109SS / 107 | MUX-2 → SE13, VX2 |

| σ kPa | total kg | full discs | half discs | trim kg |
|---|---|---|---|---|
| 5 | 4.0 | 0 | 0 | 1.25 |
| 10 | 8.0 | 1 | 0 | 0.33 |
| 15 | 12.0 | 1 | 1 | 1.88 |
| 20 | 16.0 | 2 | 1 | 0.96 |
| 25 | 20.0 | 3 | 1 | 0.04 |

How it works
Gravity gives an exactly constant surcharge; fixed part ≈ 2.8 kg ≈ 3.4 kPa (piston, carrier, rod).
No load cell, no adjustment; the frame only guides the rod and carries the LVDTs.
Weights lag the freezer air by ≈ 0.6 K at the 0.1 K/h ramp (vypocty.md §7).

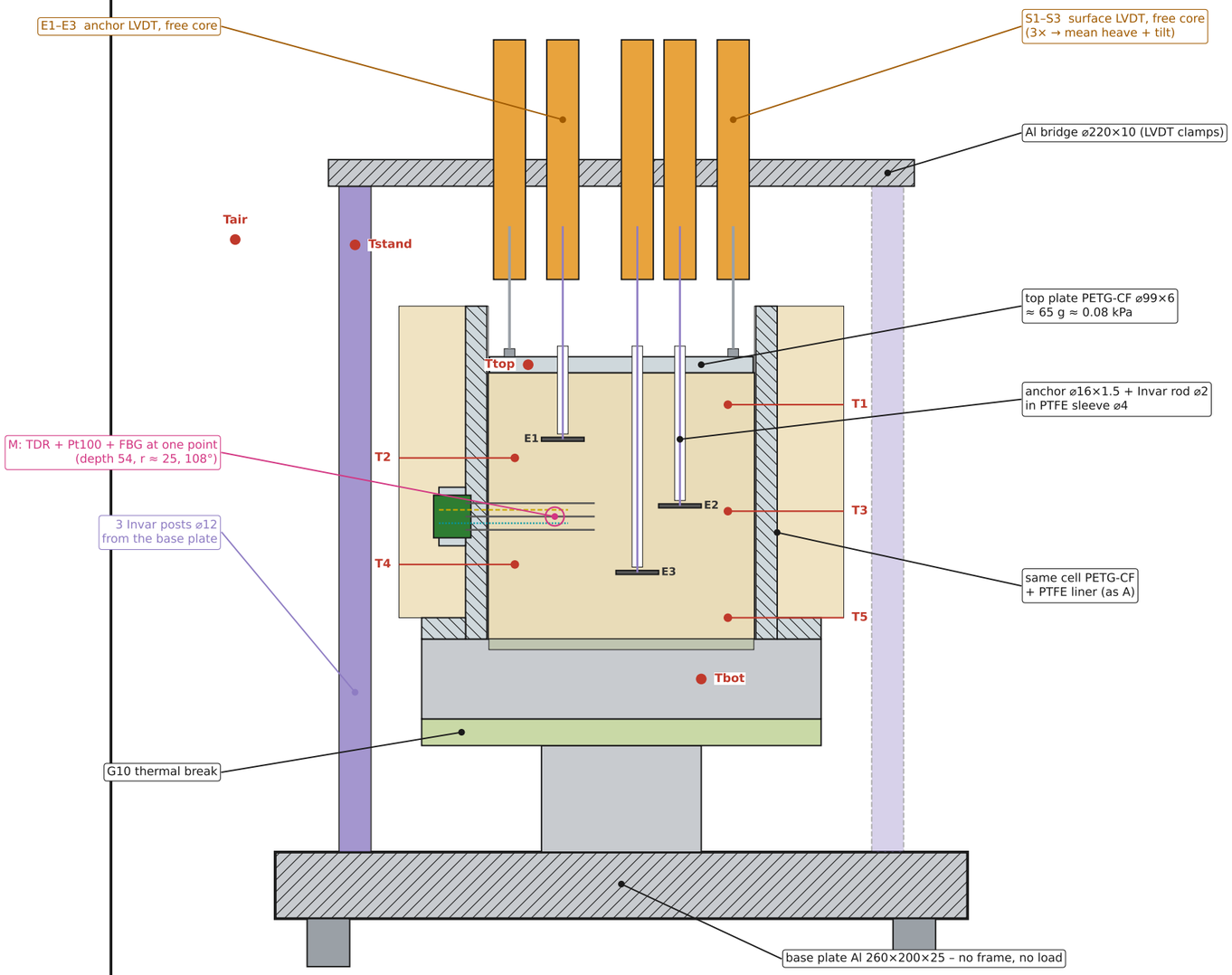
**DETAIL – sand, pressure sensors, thermistors ≈ 1 : 1.6**




schematic; orange = displacement, blue = pressure, red = temperature; cells and LVDTs revolved into section

# Variant A – unsaturated sand, dead-weight surcharge ≤ 25 kPa

| | | | | |
|---|---|---|---|---|
| status | DRAFT | drawn | Claude Code (draft) | checked: — |
| units | mm | project | E7 – freeze–thaw loading tests | date 2026-10-20 |
| scale | **≈ 1 : 3** | drawing no. | **E7-V01** | sheet 1 / 9 |



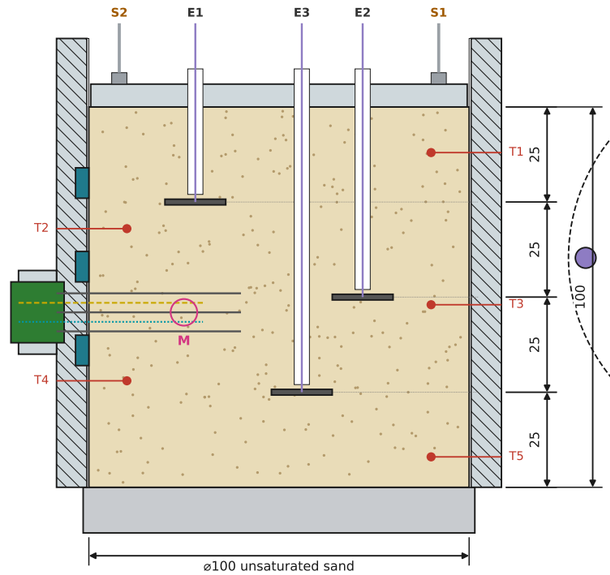

## Sensors (tag → model → CR1000)

| tag | measures | recommended model | CR1000 |
|---|---|---|---|
| S1…S3 | surface heave / settlement | RDP DCTH200, free core [U] | SE3–5 |
| E1…E3 | anchor displ. at 25/50/75 | RDP DCTH200, free core [U] | SE6–8 |
| PW1…PW3 | lateral stress (optional) | TML PDA-PA 200 kPa (flush) | MUX-1 (4×16), BrFull, VX3 2 V |
| T1…T5 | soil, r 40 mm | Campbell 109SS | MUX-2 → SE13, VX2 |
| M: TDR | water / unfrozen water | mini 3-rod TDR 40 mm [U] + TDR200 | SDM C1–C3 |
| M: Pt100 | reference temperature | Pt100 1/10 DIN, 4-wire, ø3 | MUX-1 + 4WPB100, BrHalf4W |
| M: FBG | optical temperature | FBG temp. sensor in capillary ø1 [U] | PC (Ethernet), time-synced |
| Ttop, Tbot | top plate, pedestal | Campbell 109SS | (same) |
| Tstand, Tair | Invar stand, air | Campbell 109SS / 107 | MUX-2 → SE13, VX2 |

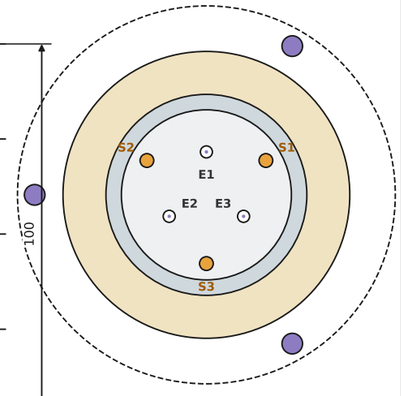
What is measured in each freeze–thaw cycle
- surface: heave during freezing, settlement during thawing, residual after the cycle (mean of S1–S3; their difference = tilt of the top plate)
- 4 layers of 25 mm: $\Delta h_1 = \bar{S} - E1$, $\Delta h_2 = E1 - E2$, $\Delta h_3 = E2 - E3$, $\Delta h_4 = E3$ (base fixed) strain $\varepsilon_i = \Delta h_i / 25$ mm, resolution ≈ $4 \cdot 10^{-5}$ (vypocty.md §3)
- frost depth from T1–T5 (0 °C isotherm) → strain vs. frozen / unfrozen layer
- after the last cycle: height (caliper), mass, water content of 5 slices × 20 mm

No load: free-core LVDTs push nothing; the plate adds 0.08 kPa only.
Reference error ≈ 50 µm per 20 K → blank run with Al dummy (vypocty.md §4).

**DETAIL – anchors and top plate ≈ 1 : 1.6**




**PLAN – top plate 1 : 3.6**

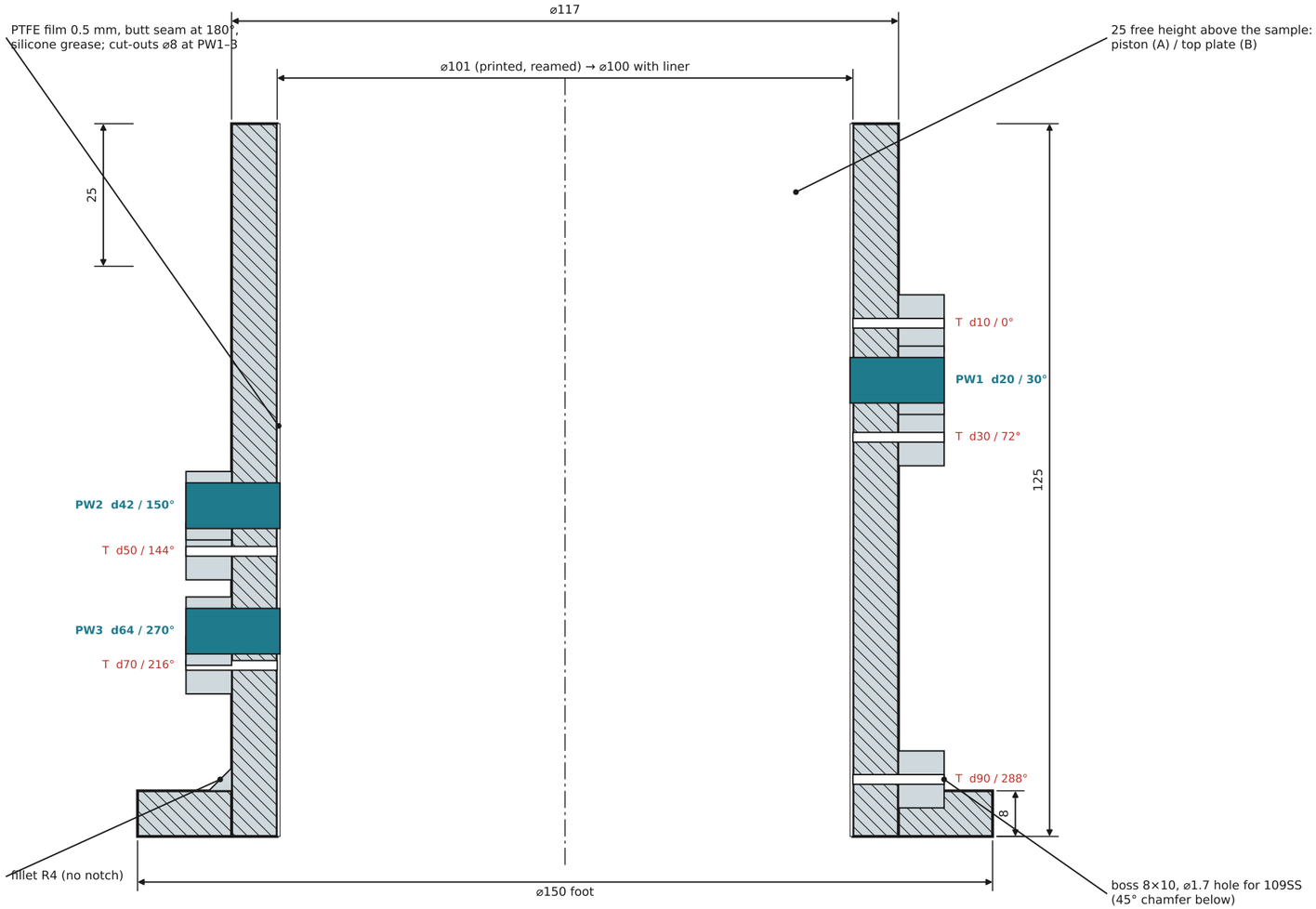



orange S = surface LVDT (r 40)
E = anchor rod hole (r 25)
violet = Invar posts (r 100)

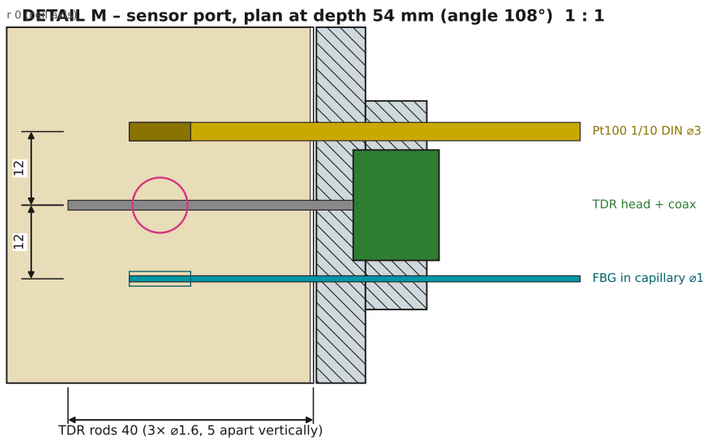
schematic; orange = displacement, red = temperature, violet = Invar reference; sensors revolved into section (true positions on the plan)

## Variant B – same sample, no load: heave, settlement, layer strain

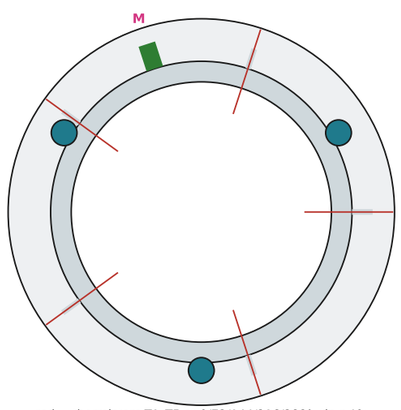
| | | | | |
|---|---|---|---|---|
| status | DRAFT | drawn | Claude Code (draft) | checked: — |
| units | mm | project | E7 – freeze–thaw loading tests | date 2026-10-20 |
| scale | **≈ 1 : 2** | drawing no. | **E7-V02** | sheet 2 / 9 |




**VIEW F – cell from above 1 : 2.4**




**DETAIL M – sensor port, plan at depth 54 mm (angle 108°) 1 : 1**




red = thermistors T1–T5 at 0/72/144/216/288°, tip r 40;
blue = PW1–3 at 30/150/270°; green = point M at 108°

## Material check – PETG-CF

| load case | stress | SF short | SF long |
|---|---|---|---|
| hoop: A 25 kPa surcharge, K0 = 1 | 0.16 MPa | > 250 | > 75 |
| hoop: B ice on the wall 200 kPa (upper est.) | 1.3 MPa | 32–44 | 9–13 |
| hoop: extreme 1 MPa | 6.3 MPa | 6–9 | 1.9–2.6 |
| axial (across layers): friction, A 25 kPa | < 0.05 MPa | > 300 | > 90 |

Assumed strength (data-sheet range, verify): XY 40–55 MPa, Z 15–30 MPa, E 3–6 GPa; long-term limit ≈ 30 % (creep, layer bond). Details: vypocty.md §5.

Verdict: suitable for both variants as a solid print (no sleeve needed).

Wall friction (the price of a solid cell, vypocty.md §5)
- A: friction carries part of the surcharge into the wall; with the PTFE liner + grease (μ ≈ 0.05) the vertical stress at the base is ≈ 90 % of the top, without the liner (μ ≈ 0.5) only ≈ 30–45 % → compare PV1 / PB1 with it.
- B: friction and ice adhesion hold back heave of the frozen top layer; PTFE has very low ice adhesion – liner and grease are mandatory.

Conditions
- wall 8 mm, concentric perimeters only, 100 % infill;
- no printed threads or sharp notches (CF lowers toughness at −10 °C);
- inside sealed with a thin epoxy coat before the liner goes in;
- cell not clamped at the foot – friction then pushes, never pulls layers.

Print settings (starting point)
hardened nozzle 0.6 mm, layer 0.2 mm, 250–260 °C, bed 80 °C, slow outer walls, filament dried 4 h at 65 °C, enclosure, part cooling ≤ 30 %.
Print upright, bosses with 45° chamfer underneath (no supports).

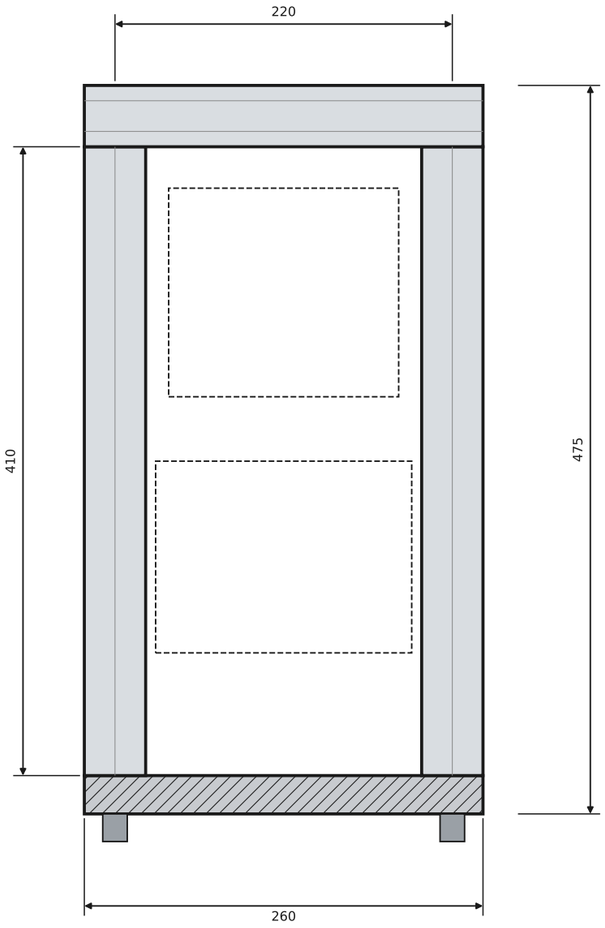
Qualification before use
1 10 freeze–thaw cycles −10/+10 °C of water-saturated coupons, then tensile test XY and Z vs. dry control;
2 bore check ø101 ± 0.1 at 3 heights, at +20 and −10 °C.

print upright in one piece; hoop direction = layer direction; foot rests on the pedestal, not bolted

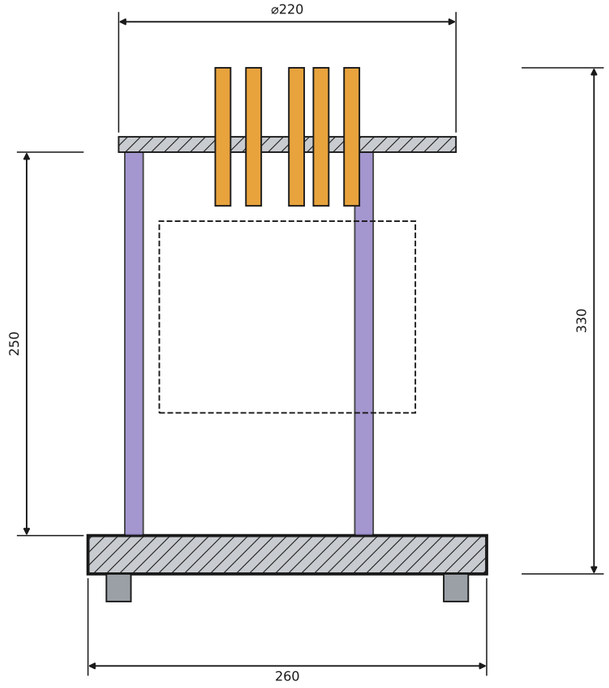
# Sample cell – PETG-CF, solid 3D print (both variants)

| | | | | |
|---|---|---|---|---|
| status | DRAFT | drawn | Claude Code (draft) | checked: — |
| units | mm | project | E7 – freeze–thaw loading tests | date 2026-10-20 |
| scale | **1 : 1** | drawing no. | **E7-V03** | sheet 3 / 9 |

## A – FRONT VIEW  1 : 4




## B – FRONT VIEW  1 : 4




## Bill of materials

| # | part | size | qty |
|---|---|---|---|
| 1 | base plate Al 6082 T6 (A and B) | 260 × 200 × 25 | 2 |
| 2 | A: profile 40×40 column | l = 410 | 2 |
| 3 | A: profile 40×40 cross-bar | l = 260 | 1 |
| 4 | A: linear ball bushing ø16 in cross-bar | | 1 |
| 5 | A: Invar post ø10 + clamp arm (LVDT) | l ≈ 300 | 2 |
| 6 | B: Invar post ø12, M6 both ends | l = 250 | 3 |
| 7 | B: Al bridge ø220×10, 6 LVDT clamps | | 1 |
| 8 | core bolt M8 / socket screw M8 | | 8 |
| 9 | levelling foot M12, steel | | 6 |

Frame A: carries nothing except the guide of the weight rod (bushing) and the LVDT posts; the surcharge goes straight down through the sample.

Stand B: three Invar posts screwed into the base plate at r 100 (60/180/300°, outside the insulation), bridge on top with split clamps for 5 LVDTs (+1 spare). The bridge is lifted off to fill the cell. Reference error of the stand ≈ 50 µm per 20 K (Al pedestal vs. Invar) → blank run with an Al dummy sample (vypocty.md §4).

Heights incl. base plate: A 475 mm (+ guide rod), B 330 mm; two rigs side by side on the tray – layout on E7-V06.

frame A only guides the weights; stand B is only a displacement reference – neither carries load

# Frame A (aluminium profiles) and Invar stand B

| | | | | |
|---|---|---|---|---|
| status | DRAFT | drawn | Claude Code (draft) | checked: — |
| units | mm | project | E7 – freeze–thaw loading tests | date 2026-10-20 |
| scale | **1 : 4** | drawing no. | **E7-V04** | sheet 4 / 9 |

## Options per measurement (PICK = recommended)

| sensor | maker / model | key specs | verdict |
|---|---|---|---|
| LVDT ±10 | RDP DCTH400 | ±5 V / 0–10 V out, −40…+80 °C, ±0.25 % FS | PICK (A) |
| LVDT ±10 | Micro-Epsilon DTA-10D + MSC7401 | 0–5 V / 4–20 mA, −20…+80 °C | alternative |
| LVDT ±10 | Solartron AX/10/S + OD5 | ±5 V out, needs 230 V conditioner | alternative |
| LVDT ±10 | HBM WA-L 20 mm | passive, needs carrier amplifier | alternative |
| LVDT free | RDP DCTH200 unguided [U] | ±5 mm, no spring, core on ø3 rod; rig B | PICK (B) |
| LVDT free | Micro-Epsilon DTA-5G8 + MSC7401 [U] | ±5 mm, free core, 0–5 V, −20…+80 °C | alternative |
| LVDT free | Solartron AX/5/F (free core) + OD5 [U] | ±5 mm, ±5 V via conditioner | alternative |
| laser | Micro-Epsilon optoNCDT 1420 | non-contact, but op. 0…+50 °C, frost on window | not suitable |
| LVDT ±10 | TE DC-EC 500 | operating only 0…+70 °C | not suitable |
| p cell | TML PDA-PA 200 kPa | ø6.5×1 mm, 0.8 mV/V, comp. −10…+60, op. −20…+70 °C | PICK |
| p cell | Kyowa BEC-A 200/500 kPa | ø30 mm, for model tests; t, temp. [U] | one cell only |
| p cell | TML KDE/KDF-PA | ø50 mm – too large for ø100 sample | not suitable |
| p wall | TE/MEAS EPL | miniature, comp. −40…+90 °C, ø [U] | alternative |
| p cell | Tekscan FlexiForce A201 | thin film, drift < 5 %/log t | comparison only |
| T probe | Campbell 109SS | ø1.6 mm, ±0.38 °C at 0 °C, direct on CR1000 | PICK |
| T probe | Campbell 107 | ø7.6 mm, ±0.2 °C (0–50 °C) | air / frame |
| T probe | TE/YSI 44031 bead | ±0.1 °C, own half bridge needed | for 0.01 °C (E6 chain) |
| Pt100 | Pt100 1/10 DIN 4-wire ø3 + 4WPB100 | ±0.03 °C at 0 °C, via MUX-1 | PICK (point M) |
| TDR | mini 3-rod 40 mm (custom) + TDR200 | fits ø100 sample; calibrate in the sand | PICK [U] |
| TDR | Campbell CS645 (3 rods 75 mm) | too long for r 50 – would cross the axis | not suitable |
| TDR | Easy Test LP/ms micro probe + own meter [U] | rods ≈ 53 mm; separate logger | alternative |
| optical | FBG in capillary ø1 + HBM FS22 / Luna si155 [U] | ≈ ±0.2–0.5 °C, many points on one fibre | PICK [U] |
| optical | fluorescent probe Opsens OTG-F + TempSens [U] | ±0.3 °C, analog out → CR1000 SE | alternative |

## CR1000 channel plan

| terminal | connected to | measurement |
|---|---|---|
| SE1–SE2 | LVDT-A1, A2 (rig A) | VoltSE ±5000 mV, averaging; own ±15 V supply |
| SE3–SE5 | LVDT S1–S3 (rig B surface) | VoltSE ±5000 mV |
| SE6–SE8 | LVDT E1–E3 (rig B anchors) | VoltSE ±5000 mV |
| DIFF5, DIFF6 | MUX-1 AM16/32B, 4×16 | 9 p-bridges + 2 Pt100 (4WPB100); COM H2/L2 → DIFF5 |
| VX3 | MUX-1 excitation | 2000 mV bridges / 2500 mV Pt100 |
| C1–C3 | TDR200 (SDM) → SDMX50SP | 2 TDR probes (point M, A and B) |
| C4, C5 | MUX-1 reset, clock | |
| SE13 | MUX-2 AM16/32B, 2×32 | ≈ 20 thermistors (Therm109), VX2 |
| C6, C7 | MUX-2 reset, clock | |
| 12V, G | both multiplexers, TDR200 | |
| – | FBG interrogator | 1 fibre for both rigs; own PC, clock synced |
| SE14–16 | spare | |

Channel count (2 rigs): 8 LVDTs, 9 pressure bridges, ≈ 20 thermistors,
point M ×2: 2 TDR (TDR200 + SDMX50SP), 2 Pt100 (MUX-1), 2 FBG (interrogator).
No load cells: A is loaded by dead weight, B is not loaded at all.

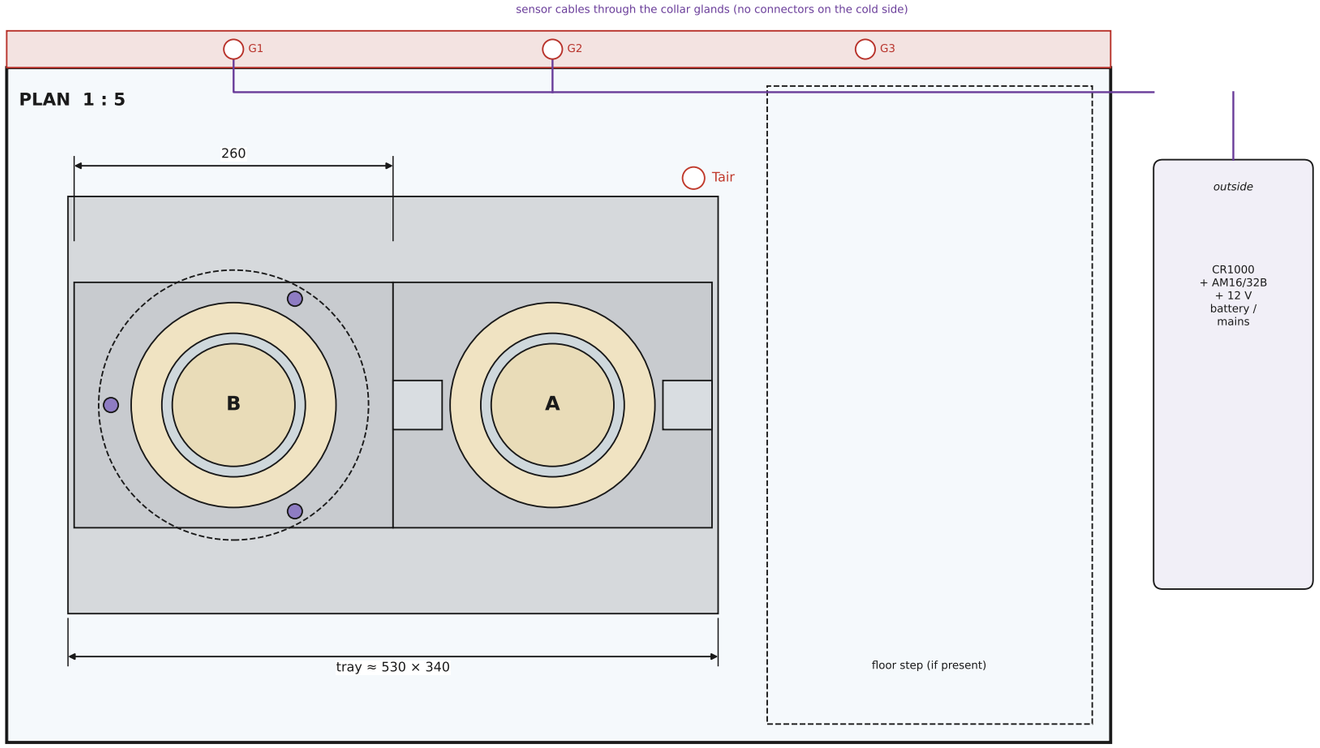
Resolution with the CR1000 (13-bit ADC, ±5000 mV range)
- LVDT ±10 mm (A): ≈ 1.3 µm; LVDT ±5 mm (B): ≈ 0.7 µm
  → layer strain over 25 mm ≈ $4 \cdot 10^{-5}$ (B)
- absolute linearity ±0.25 % FS → calibrate each LVDT with a micrometer
- 109SS: ±0.38 °C at 0 °C; ice-point offset check improves it

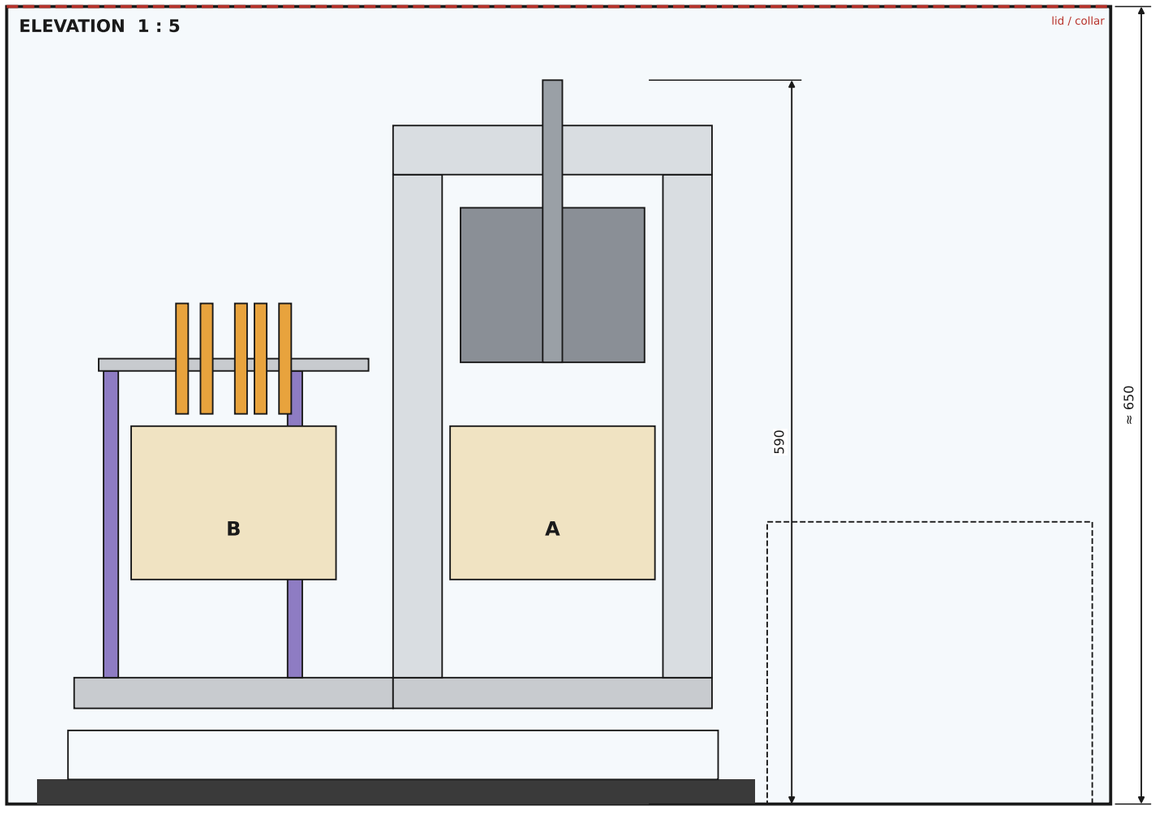
Notes
- Rig B uses LVDTs with a FREE (unguided) core – no spring force on the sample; the core rides on the anchor rod / plate rod. Check the free-core version and the core length with RDP before ordering [U].
- DC-LVDTs need their own ±15 V (or 24 V) supply, logger only reads ±5 V.
- CR1000 is discontinued (successor CR1000X / CR1000Xe, 24-bit); the existing one is fine for these resolutions.
- Logger outside the freezer (CR1000 −25…+50 °C).
- Point M: TDR, Pt100 and FBG sit 12 mm apart at one depth – Pt100 is the reference for the FBG and for T3; TDR gives unfrozen water vs. temperature.
- Pressure cells: calibrate in the same sand under dead weight (5–25 kPa) at +10 and −10 °C; correct zero(T) with the neighbouring thermistor.

both rigs on one CR1000 + AM16/32B; models from manufacturer data (see README §7)

# Sensor selection and CR1000 wiring

| | | | | |
|---|---|---|---|---|
| status | DRAFT | drawn | Claude Code (draft) | checked: — |
| units | mm | project | E7 – freeze–thaw loading tests | date 2026-10-20 |
| scale | – | drawing no. | **E7-V05** | sheet 5 / 9 |




## Fit check

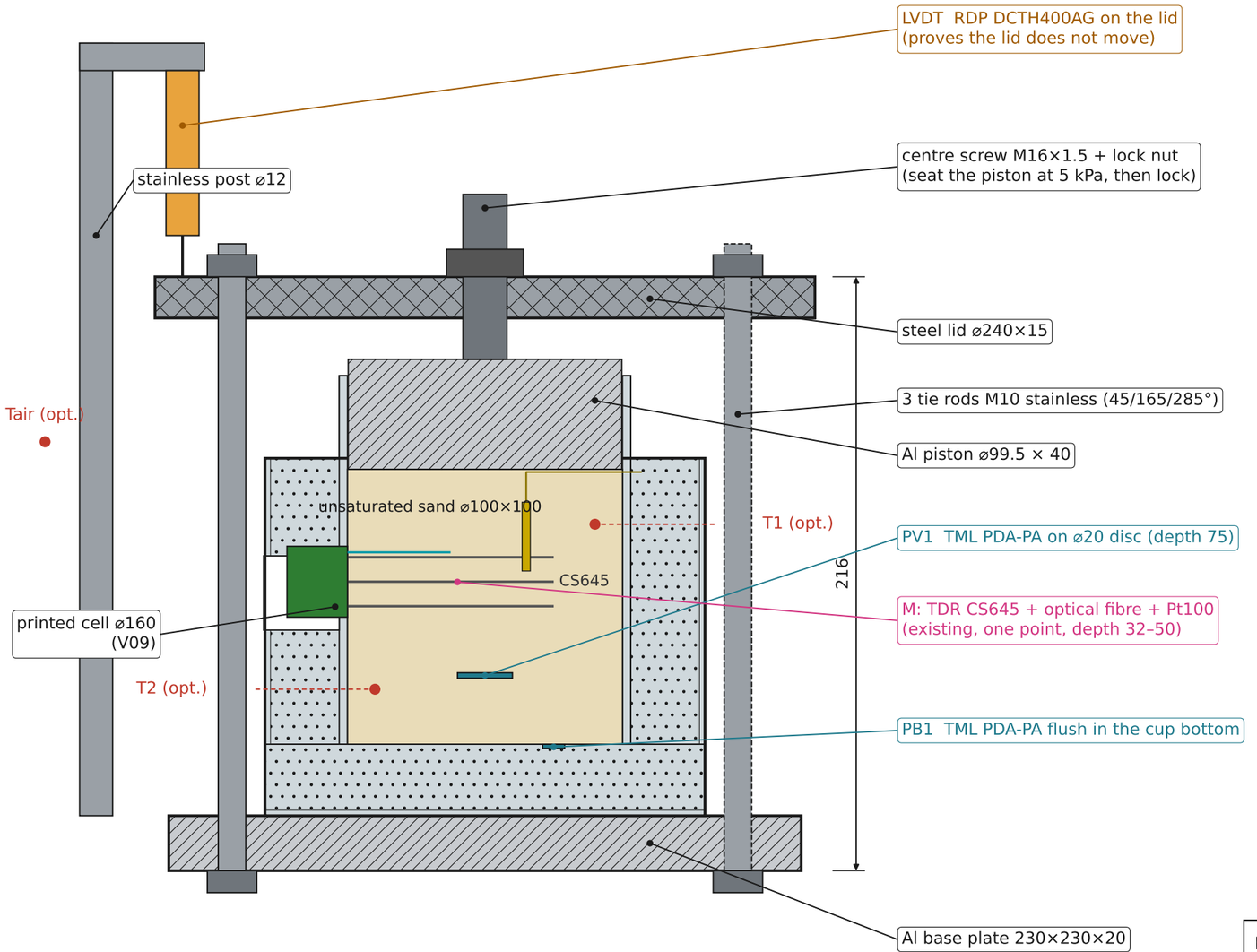
| item | size | limit | status |
|---|---|---|---|
| base plates 2 × 260 × 200 | 520 × 200 | tray ≈ 530 × 340 | OK (tight) |
| rig A (weights) | 20 kg steel discs ø150 | nothing outside frame | OK |
| rig B (stand) | 408 mm high | no load, no frame | OK |
| highest point | 590 mm | inside ≈ 650 | OK, measure |
| logger | outside, 0…+40 °C | CR1000 −25…+50 °C | keep warm & dry |

Notes
- Both rigs share the freezer program (Set A) and the CR1000; each rig has its own Tair sensor at sample mid-height.
- Same sand, same printed cell, same thermistors in A and B – the only difference is the load (A ≤ 25 kPa dead weight, B none).
- Run the blank test with Al dummies in both rigs before the first sand run.
- Dimensions marked ≈ are estimates from the photos (E6) – please measure.

same freezer as E6 (inside ≈ 900 × 550 × 650, to be measured); logger outside

**Both rigs in the chest freezer + CR1000**

| | | | | |
|---|---|---|---|---|
| status | DRAFT | drawn | Claude Code (draft) | checked: — |
| units | mm | project | E7 – freeze–thaw loading tests | date 2026-10-20 |
| scale | **1 : 5** | drawing no. | **E7-V06** | sheet 6 / 9 |




## To buy – type, quantity, price (EUR excl. VAT, ≈)

| tag | type | pcs | EUR | CR1000 |
|---|---|---|---|---|
| PV1, PB1 | TML PDA-PA 500 kPa (ø6.5 × 1 mm) [U] | 2 | 1 200–2 000 | DIFF1, DIFF2 BrFull, VX1 |
| LVDT | RDP DCTH400AG (±10 mm, spring) | 1 | 581 * | SE9 |
| + | DIN-rail power supply for the LVDT | 1 | 40–80 | – |
| T1, T2, Tair | Campbell 109SS-L (ø1.6 mm) | 3 | 540–780 | SE11–13, VX3 |

sensors ≈ 2 361–3 441 EUR  mechanics ≈ 290–590 EUR → total ≈ 2 651–4 031 EUR
* price found at a dealer (Oct 2026, see README); others = estimates → ask for quotes.
Already available (no cost): TDR CS645, Pt100, optical fibre (point M), CR1000.

## Existing sensors at point M

| tag | type | connection |
|---|---|---|
| TDR | Campbell CS645, 3 rods ø3.2 × 75, pitch 8.9 | TDR100/200, SDM |
| Pt100 | Pt100, 37 mm (25 mm metal), vertical | 4-wire |
| OF | optical fibre, tip 37.5 from the wall | own interrogator |

Fixed lid (vypocty.md §9)
• piston seated by the centre screw at 5 kPa, locked → the sample cannot heave;
  freezing pressure is read by PV1 (in the sand) and PB1 (at the bottom)
• at 1 MPa: tie rods +49 µm, lid 60 µm → piston moves ≈ 0.1 mm (≈ 20 µm at 200 kPa)
• stainless tie rods and the inner stack shrink almost equally (Δ ≈ 12 µm per 20 K)

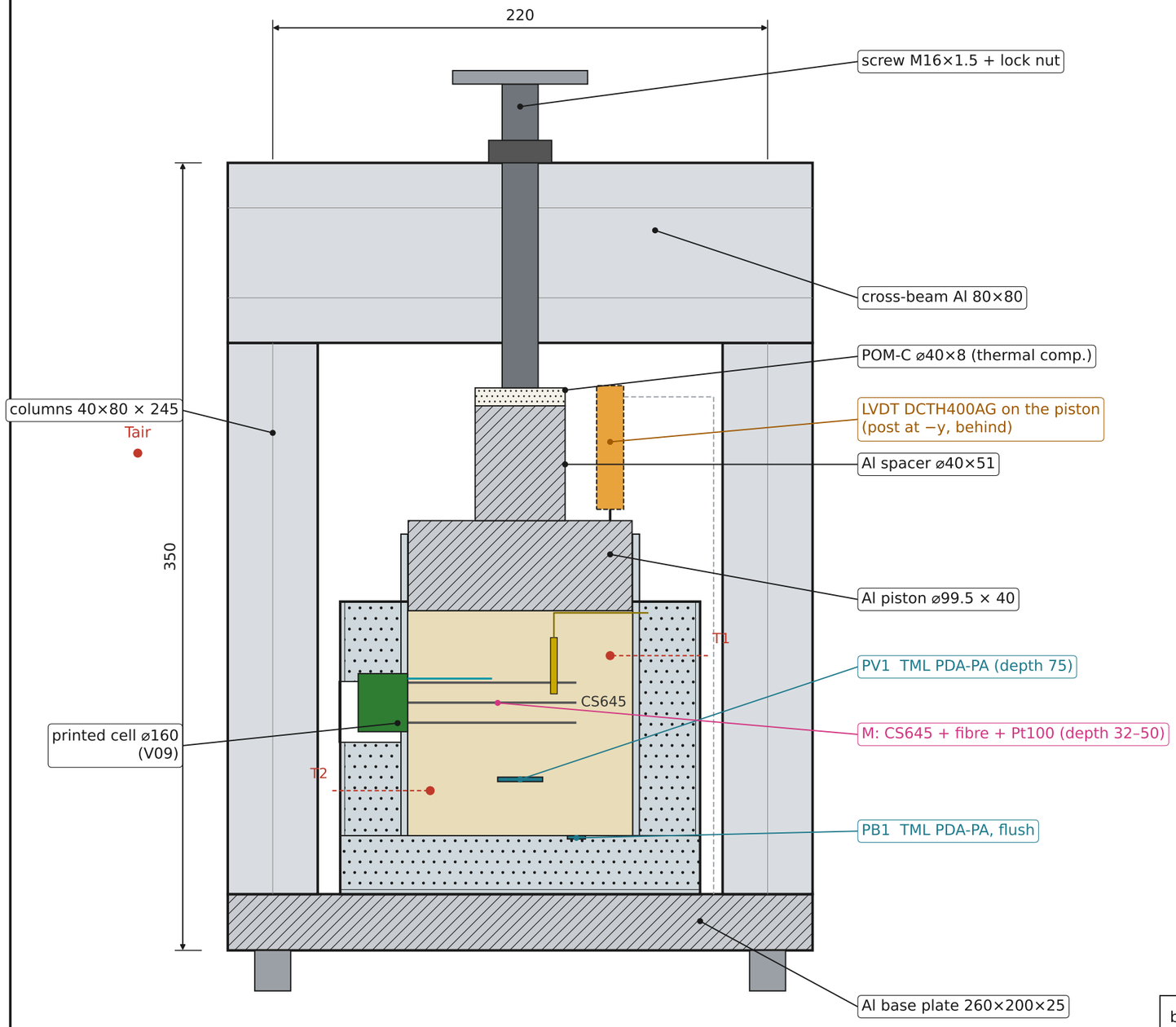
CR1000 wiring – no multiplexer
DIFF1 PV1, DIFF2 PB1 (BrFull, VX1) · SE9 LVDT · SE11–13 T1, T2, Tair (optional)
point M: per the existing devices (TDR on SDM C1–C3, temperature / fibre own reader)

Removed vs. variant A: frame, weights, carrier, ball, Invar posts, 2nd LVDT,
4 of 6 pressure cells, porous stones, pedestal, both multiplexers.

blue = pressure, red = temperature, magenta = point M

# Variant A-min – fixed lid on 3 tie rods

| | | | | |
|---|---|---|---|---|
| status | DRAFT | drawn | Claude Code (draft) | checked: — |
| units | mm | project | E7 – freeze–thaw loading tests | date 2026-10-20 |
| scale | **≈ 1 : 2.2** | drawing no. | **E7-V07** | sheet 7 / 9 |




## A-frame vs. A-min (tie rods)

| | A-frame (this sheet) | A-min (E7-V07) |
|---|---|---|
| holds the piston | frame 40×80 + 80×80, screw | steel lid on 3 tie rods M10 |
| piston movement at 1 MPa | ≈ 0.08–0.15 mm | ≈ 0.11 mm |
| thermal error (20 K) | ≈ 0 with POM-C insert | ≈ 12 µm |
| access to the sample | open sides, screw from top | lid must be removed |
| mechanics, EUR (≈) | ≈ 375–760 | ≈ 330–670 |
| sensors to buy, EUR (≈) | 2 360–3 440 (same) | 2 360–3 440 |

## To buy (same sensors as A-min)

| tag | type | pcs | EUR | CR1000 |
|---|---|---|---|---|
| PV1, PB1 | TML PDA-PA 500 kPa (ø6.5 × 1 mm) [U] | 2 | 1 200–2 000 | DIFF1, DIFF2 BrFull, VX1 |
| LVDT | RDP DCTH400AG (±10 mm, spring) | 1 | 581 * | SE9 |
| + | DIN-rail power supply for the LVDT | 1 | 40–80 | – |
| T1, T2, Tair | Campbell 109SS-L (ø1.6 mm) | 3 | 540–780 | SE11–13, VX3 |

* price found at a dealer (Oct 2026); others estimated – ask for quotes.
Already available: TDR CS645 + reflectometer, Pt100, optical fibre (point M), CR1000.

How it works
- turn the screw until PV1 reads ≈ 5 kPa, lock the nut – the piston cannot move up;
- freezing pressure is read by PV1 (in the sand) and PB1 (bottom); LVDT proves the piston stays put (≈ 0.1 mm at 1 MPa, joints of the profiles dominate);
- Al columns shrink more than the sand stack – the POM-C insert expands the stack back (Δ ≈ 1 µm per 20 K), blank run with an Al dummy confirms it.

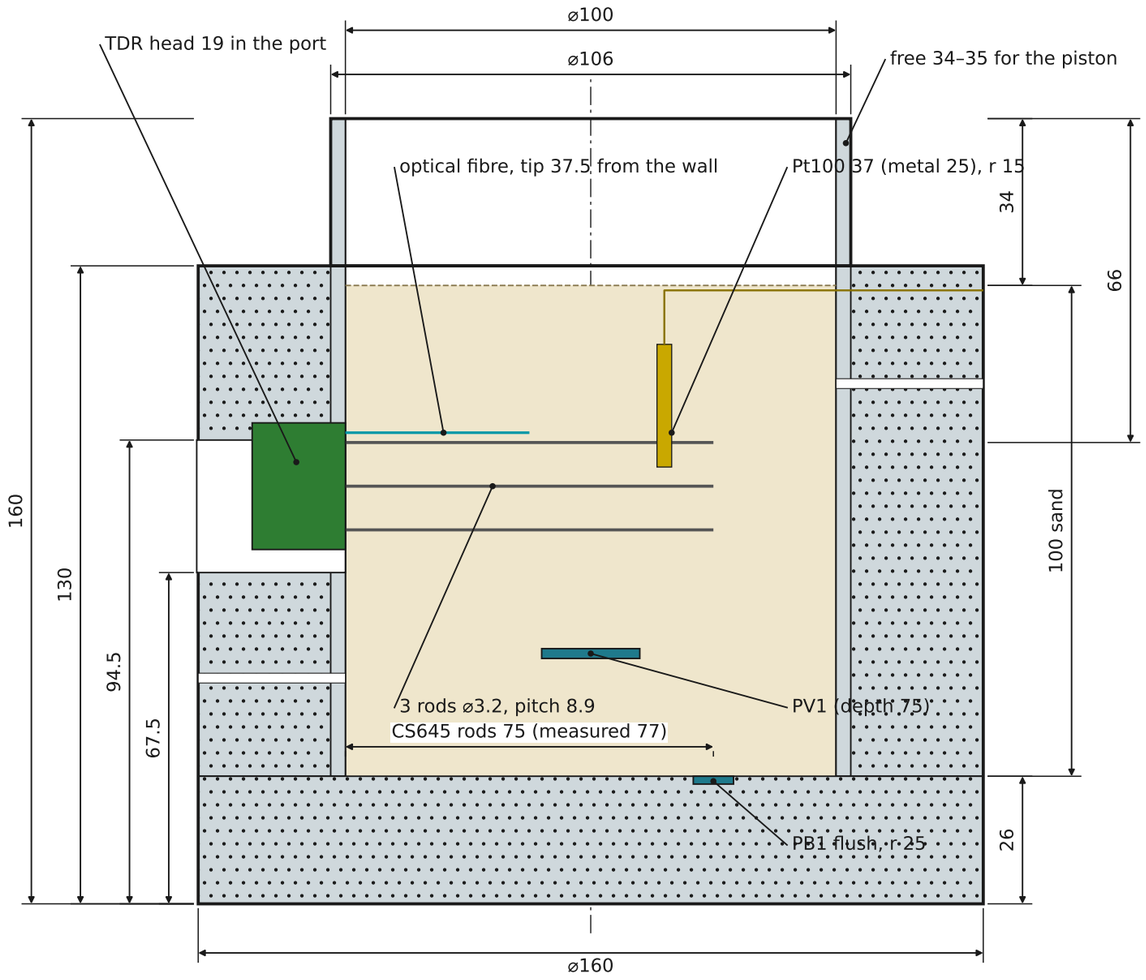
Details: vypocty.md §10.

blue = pressure, red = temperature, magenta = point M; dashed = behind

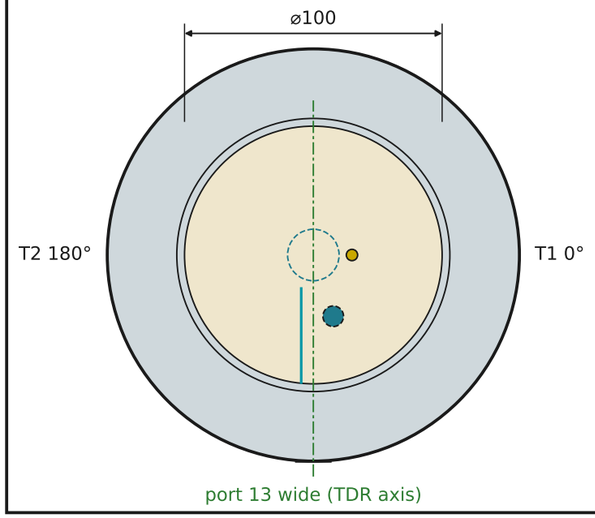
# Variant A-frame – piston fixed by an Al frame

| | | | | |
|---|---|---|---|---|
| status | DRAFT | drawn | Claude Code (draft) | checked: — |
| units | mm | project | E7 – freeze–thaw loading tests | date 2026-10-20 |
| scale | **≈ 1 : 2.2** | drawing no. | **E7-V08** | sheet 8 / 9 |

## SECTION A–A  1 : 1.25 (port and sensors revolved)




## PLAN  1 : 2.4




## Sensors at point M (existing)

| | sensor | position |
|---|---|---|
| TDR | Campbell CS645 | rods 75 (77 meas.), pitch 8.9; top rod 66 below the collar top |
| OF | optical fibre | horizontal at the top rod, tip 37.5 from the wall |
| Pt100 | 37 mm (25 metal) | vertical from the sand surface, 15 from the axis |

Depths below the sand surface (z = 126): rods 32 / 41 / 50,
fibre ≈ 31, Pt100 metal 12–37 → all three sense the 30–50 mm layer.
Open (sketch 7.7.26): does the Pt100 measure at the tip or along its length?

Cable routes (to print or drill)
- TDR head + coax and the fibre: through the side port 13 × 27
- Pt100 cable: groove under the piston, ⌀4 hole at z ≈ 125 (0°)
- PV1: ⌀3.5 hole at z = 51 (90°); PB1: groove in the cavity floor + ⌀3.5 hole through the jacket at 288°
- optional T1 (depth 20, 0°), T2 (depth 80, 180°): ⌀2 holes

Print (PrusaSlicer, MK4S): PETG(-CF), 0.2 mm, ≈ 600 g, 21 h
- jacket 10 % infill = insulation (replaces Kaiflex)
- recommended: 100 % infill under the cavity (z 0–26) – the bottom carries the full freezing pressure (10 % gyroid crushes at ≈ 1–2 MPa)
- wall ≈ 5 mm (≈ 8 perimeters) around the cavity: 3 mm alone sees ≈ 3 MPa hoop stress at 200 kPa, ≈ 17 MPa at 1 MPa
- epoxy coat + PTFE spray + silicone grease inside the cavity

per drawing M. Vrána 23.06.2026 + sensor sketch; z = 0 at the underside

# Cell A – printed body ⌀160 (Column_one_body_version_1)

| | | | | |
|---|---|---|---|---|
| status | DRAFT | drawn | Claude Code (draft) | checked: — |
| units | mm | project | E7 – freeze–thaw loading tests | date 2026-10-20 |
| scale | **1 : 1.25** | drawing no. | **E7-V09** | sheet 9 / 9 |